

UCHECHUKWU ONWUKA

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HIGHLIGHTS OF QUALIFICATIONS

- Extensive experience in modeling, design, and optimization of robotic systems, with a focus on control systems.
- Proficient in MATLAB, Python, C++, ROS, and advanced computer-aided design software (SOLIDWORKS, AutoCAD).
- Strong academic writing and research skills, with a publication in the International Journal of Robotics and Automation.
- Proven ability to conduct failure mode and effects analysis (FMEA) and implement real-time path planning algorithms.
- Excellent teamwork, problem-solving, and communication skills, honed through research and teaching experience.

EDUCATION

Abia State University, Nigeria

Sep. 2018 – Sep. 2022

Bachelor of Engineering. Mechanical Engineering

Relevant Coursework: Control Systems, Robotics, Thermal-Fluid Sciences, Mechanical Design

Thesis: Development and Optimization of a Cassava Peeling Machine

SKILLS

- **Programming Skills:** MATLAB, Python, ROS, C++, R, Rstudio, VS-code, advanced Excel, PowerBI, OpenCV, TensorFlow.
- **Design and Configurations:** SOLIDWORKS, V-REP, Gazebo, Autodesk AutoCAD, fusion 360, Proteus
- **Algorithms:** SLAM, Path Planning, Kalman Filters, Machine Learning
- **Soft Skills:** Teamwork, Problem-Solving, Communication, Leadership.

EXPERIENCE

Research Assistant

Abia State University

Jan 2021 -- Jan 2022

- Participated in modeling and computational analysis of robotic manipulators using MATLAB and ROS (Robot Operating System).
- Designed and simulated the kinematics and dynamics of a mobile robot platform using SOLIDWORKS and V-REP.
- Assisted in academic writing and reviewing journals for my research group before publications.
- Conducted literature review on robotic systems and identified research gaps for future studies.

WORK EXPERIENCE

Robotics Engineer

Cassey Dil Mechatronics Centre.

Apr 2023—Present

- Performed failure mode and effects analysis (FMEA) on robotic systems and provided recommendations to the management team.
- Prepared detailed cost analysis, technical documentation, and maintenance reports related to the deployment, operation, and servicing of industrial robotic arms, automated guided vehicles (AGVs), and collaborative robots (cobots) at client manufacturing facilities.

- Developed a vision-based robotic grasping system using CNNs in ROS, improving object handling success rate by 18%.
- Implemented a real-time path planning algorithm using A* search in C++ combined with dynamic obstacle avoidance in ROS.
- Developed a distributed control strategy for leader-follower formation control in multi-robot systems using ROS.

Engineering Intern

Omotosho Power Plant

May 2022 -- Feb 2023

- Performed condition-based maintenance on gas turbine combustion components and hot gas path parts to evaluate their performance for the next maintenance cycle.
- Conducted a complete overhaul of the turbine air inlet compartment and inspected the air plenum during installation, startup, and commissioning processes.
- Carried out non-destructive testing (NDT) on turbine spare parts, such as shroud blocks, turbine blades, and combustion liners.
- Prepared a detailed report documenting all the technical activities performed during the outage, serving as feedback and future reference for the management team.
- Worked on implementing and testing vision-based navigation systems for autonomous robots that were used for troubleshooting on the gas turbines.

TEACHING EXPERIENCE

Programming Instructor

Government Technical College

Jan 2021 – April 2022

- Prepared lecture notes and introduced 50 grade 8 students to the fundamentals of robotics and programming.
- Organized interactive sessions and hands-on lab activities for the students to build and program basic robots.
- Developed assessment rubrics for midterms and final exams, and provided feedback to help students improve their understanding of robotics concepts.

PUBLICATIONS & RESEARCH

- Undergraduate Thesis: "Development and Optimization of a cassava peeling machine"
 - Nwadinobi .C.P, **Onwuka A.U** "Development and Optimization of a cassava Peeling Machine*," International Journal of Robotics and Automation (IJRA), vol. 8, no. 2, pp. 45-56, 2023.
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Academic Awards & Professional Membership

- Best Graduating Student in Mechanical Engineering
- Founders Award for Academic Excellence
- Best Project in Robotics and Automation
- Institute of Electrical and Electronics Engineers (IEEE): Student Member